SDIS 39

Installation et configuration du bastion d’administration « Apache Guacamole »

GSIC

Thibault BAILLY
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PREREQUIS

- Ubuntu Server 24.04 LTS
- Accès administrateur
- Une IP réservée
- DNS internes
- Certificat SSL (guacamole.sdis39.fr)
- Attribution nom DNS

CONFIGURATION RESEAU (NETPLAN)

EDITER LE FICHIER NETPLAN

```
~$ sudo nano /etc/netplan/00-installer-config.yaml
```

Collé le contenu ci-dessous :

```
network:
  version: 2
  renderer: networkd
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 172.16.36.41/24 #IP du Guacamole
      gateway4: 172.16.36.254
      nameservers:
        addresses:
          - 192.168.0.21
          - 192.168.0.22
```

Appliquer la configuration

```
~$ sudo netplan apply
```

INSTALLATION DE DOCKER

INSTALLER LES DEPENDANCES :

```
~$ sudo apt update && sudo apt upgrade -y
~$ sudo apt install -y ca-certificates curl gnupg lsb-release
```

AJOUTER LE DEPOT DOCKER

```
~$ sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --
dearmor -o /etc/apt/keyrings/docker.gpg
echo \
"deb [arch=$(dpkg --print-architecture) signed-
by=/etc/apt/keyrings/docker.gpg] \
 https://download.docker.com/linux/ubuntu \
 $(lsb_release -cs) stable" \
| sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

INSTALLER DOCKER

```
~$ sudo apt update
~$ sudo apt install -y docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

AJOUTER L'UTILISATEUR AU GROUPE DOCKER

```
~$ sudo usermod -aG docker $USER
```

Reconnecte-toi :

```
~$ exit
```

Test :

```
~$ docker run hello-world
```

INSTALLATION DE GUACAMOLE VIA DOCKER

CREATION DE L'ARBORESCENSE

```
~$ sudo mkdir -p /var/docker/guacamole
~$ sudo mkdir -p /var/docker/guacamole/initdb
~$ sudo mkdir -p /var/docker/guacamole/config
~$ sudo mkdir -p /var/docker/guacamole/ldap
~$ sudo mkdir -p /var/docker/guacamole/recordings

~$ sudo chown -R $USER:$USER /var/docker/guacamole
```

GENERATION DU SCHEMA SQL GUACAMOLE POSTGRESQL

```
~$ cd /opt/docker/guacamole
~$ mkdir initdb
~$ docker run --rm guacamole/guacamole:1.6.0 \ /opt/guacamole/bin/initdb.sh --postgresql > /var/docker/guacamole/initdb/initdb.sql
```

CREATION DU FICHIER DOCKER-COMPOSE.YML

```
~$ sudo nano /var/docker/guacamole/docker-compose.yml
```

```
services:
  guacd:
    image: guacamole/guacd:1.6.0
    container_name: guacd
    restart: unless-stopped
    volumes:
      - /var/lib/guacamole/recordings:/var/lib/guacamole/recordings
  postgres:
    image: postgres:17
    container_name: guac-postgres
    restart: unless-stopped
    environment:
      POSTGRES_DB: nom_BDD
      POSTGRES_USER: Nom_Utilisateur
      POSTGRES_PASSWORD: Mot_de_passe_format "*****"
      PGDATA: /var/lib/postgresql/data/guacamole
    volumes:
      - guac-db:/var/lib/postgresql/data
      - ./initdb:/docker-entrypoint-initdb.d:ro
    ports:
```

```
      - "5432:5432"
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U nom_utilisateur -d nom_BDD"]
      interval: 10s
      timeout: 5s
      retries: 5
  guacamole:
    image: guacamole/guacamole:1.6.0
    container_name: guacamole
    restart: unless-stopped
    depends_on:
      postgres:
        condition: service_healthy
      guacd:
        condition: service_started
    environment:
      # Connexion Guacd
      GUACD_HOSTNAME: guacd
      # Connexion DB
      POSTGRESQL_HOSTNAME: postgres
      POSTGRESQL_DATABASE: nom_BDD
      POSTGRESQL_USERNAME: nom_utilisateur
      POSTGRESQL_PASSWORD: "Mot_de_passe"
      #TOTP
      TOTP_ENABLED: "true"
      TOTP_ISSUER: "SDIS39"
      TOTP_DIGITS: "6"
      TOTP_PERIOD: "30"
      # LDAPS — truststore JKS passé à la JVM
      JAVA_OPTS: "-Djavax.net.ssl.trustStore=/etc/guacamole/certs/ldap-
truststore.jks -Djavax.net.ssl.trustStorePassword=changeit"
      # Configuration LDAP
      LDAP_HOSTNAME: "ton_controler_de_domaine"
      LDAP_PORT: "636"
      LDAP_ENCRYPTION_METHOD: "ssl"
      LDAP_SEARCH_BIND_DN:
"CN=svc_guacamole,OU=Connecteurs,OU=Comptes_bastions_Guacamole,OU=SDIS39,DC
=sdis39,DC=fr"
      LDAP_SEARCH_BIND_PASSWORD: "Mot_de_passe"
      LDAP_USER_BASE_DN: "OU=SDIS39,DC=sdis39,DC=fr"
      LDAP_USERNAME_ATTRIBUTE: "sAMAccountName"
      LDAP_USER_SEARCH_FILTER:
"(&(objectClass=user)(|(memberOf:1.2.840.113556.1.4.1941:=CN=g_Guacamole_gs
ic,OU=Acces_bastions,OU=Comptes_bastions_Guacamole,OU=SDIS39,DC=sdis39,DC=f
r)(memberOf:1.2.840.113556.1.4.1941:=CN=g_Guacamole_rh,OU=Acces_bastions,OU
=Comptes_bastions_Guacamole,OU=SDIS39,DC=sdis39,DC=fr)))"
      # Enregistrements de sessions
      RECORDING_SEARCH_PATH: "/var/lib/guacamole/recordings"
      GUACAMOLE_LOG_LEVEL: "debug"
    ports:
      - "8080:8080"
    volumes:
      - /var/lib/guacamole/recordings:/var/lib/guacamole/recordings
      - /var/docker/guacamole/ldap:/etc/guacamole/ldap:ro
      - /var/docker/guacamole/certs:/etc/guacamole/certs:ro
volumes:
  guac-db:
```

LANCEMENT DES CONTENEURS

```
~$ cd /var/docker/guacamole
~$ docker compose up -d
```

INSTALLATION DU REVERSE PROXY NGINX

INSTALLER NGINX

```
sudo apt install nginx -y
```

CREER LA CONFIGURATION GUACAMOLE

```
sudo nano /etc/nginx/sites-available/guacamole.conf
```

Contenu

```
server {
    listen 80;
    server_name guacamole.sdis39.fr;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    server_name guacamole.sdis39.fr;

    ssl_certificate     /etc/ssl/certs/guacamole.sdis39.fr.crt;
    ssl_certificate_key /etc/ssl/private/guacamole.sdis39.fr.key;

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers on;

    location / {
        proxy_pass http://127.0.0.1:8080/guacamole/;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto https;
        proxy_set_header X-Forwarded-Host $host;
        proxy_set_header Host $host;
        proxy_buffering off;
    }
}
```

Activer le site :

```
sudo ln -s /etc/nginx/sites-available/guacamole.conf /etc/nginx/sites-
enabled/
```

Vérifier :

```
Sudo nginx -t
```

Relancer :

```
sudo systemctl restart nginx
```

ACCES AU PORTAIL GUACAMOLE

Avec Nginx (production) :
`https://guacamole.sdis39.fr/`